

DDL

1)

```
MySQL 8.0 Command Line Client
-> MobileNo NUMBER(10),
-> City VARCHAR(15)
-> );
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '{
  RegNo NUMBER(3),
  Name VARCHAR(15),
  Gender CHAR(1),
  DOB DATE,
' at line 2
mysql> CREATE TABLE STUDENT
-> (
->   RegNo INT(3) PRIMARY KEY,
->   Name VARCHAR(15),
->   Gender CHAR(1),
->   DOB DATE,
->   MobileNo BIGINT,
->   City VARCHAR(15)
-> );
Query OK, 0 rows affected, 1 warning (0.05 sec)

mysql> desc student;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| RegNo | int  | NO   | PRI | NULL    |       |
| Name  | varchar(15) | YES |     | NULL    |       |
| Gender | char(1) | YES |     | NULL    |       |
| DOB   | date  | YES |     | NULL    |       |
| MobileNo | bigint | YES |     | NULL    |       |
| City  | varchar(15) | YES |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.01 sec)

mysql>
```

2)

```
MySQL 8.0 Command Line Client
+-----+-----+-----+-----+-----+-----+
| DOB   | date  | YES |     | NULL    |       |
| MobileNo | bigint | YES |     | NULL    |       |
| City  | varchar(15) | YES |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.01 sec)

mysql> CREATE TABLE FACULTY
-> (
->   FacNo VARCHAR(4) PRIMARY KEY,
->   FacName VARCHAR(15),
->   Gender CHAR(1),
->   DOB DATE,
->   DOJ DATE,
->   MobileNo BIGINT
-> );
Query OK, 0 rows affected (0.03 sec)

mysql> desc faculty
-> ;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| FacNo | varchar(4) | NO   | PRI | NULL    |       |
| FacName | varchar(15) | YES |     | NULL    |       |
| Gender | char(1) | YES |     | NULL    |       |
| DOB   | date  | YES |     | NULL    |       |
| DOJ   | date  | YES |     | NULL    |       |
| MobileNo | bigint | YES |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)

mysql>
```

3)

```
MySQL 8.0 Command Line Client
-> MobileNo BIGINT
-> );
Query OK, 0 rows affected (0.03 sec)

mysql> desc faculty
-> ;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| FacNo | varchar(4) | NO | PRI | NULL | |
| FacName | varchar(15) | YES | | NULL | |
| Gender | char(1) | YES | | NULL | |
| DOB | date | YES | | NULL | |
| DOJ | date | YES | | NULL | |
| MobileNo | bigint | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)

mysql> CREATE TABLE DEPARTMENT
-> (
-> DeptNo VARCHAR(4) PRIMARY KEY,
-> DeptName VARCHAR(15),
-> DeptHead VARCHAR(4)
-> );
Query OK, 0 rows affected (0.03 sec)

mysql> desc department;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| DeptNo | varchar(4) | NO | PRI | NULL | |
| DeptName | varchar(15) | YES | | NULL | |
| DeptHead | varchar(4) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> _
```

4)

```
MySQL 8.0 Command Line Client
Query OK, 0 rows affected (0.03 sec)

mysql> desc department;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| DeptNo | varchar(4) | NO | PRI | NULL | |
| DeptName | varchar(15) | YES | | NULL | |
| DeptHead | varchar(4) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> CREATE TABLE COURSE
-> (
-> CourseNo VARCHAR(3) PRIMARY KEY,
-> CourseDesc VARCHAR(14),
-> CourseType CHAR(1),
-> SemNo CHAR(1),
-> HallNo VARCHAR(4),
-> FacNo VARCHAR(4)
-> );
Query OK, 0 rows affected (0.03 sec)

mysql> desc course;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| CourseNo | varchar(3) | NO | PRI | NULL | |
| CourseDesc | varchar(14) | YES | | NULL | |
| CourseType | char(1) | YES | | NULL | |
| SemNo | char(1) | YES | | NULL | |
| HallNo | varchar(4) | YES | | NULL | |
| FacNo | varchar(4) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)

mysql>
```

5)

```
MySQL 8.0 Command Line Client
-> );
Query OK, 0 rows affected (0.03 sec)

mysql> desc course;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| CourseNo | varchar(3) | NO | PRI | NULL | |
| CourseDesc | varchar(14) | YES | | NULL | |
| CourseType | char(1) | YES | | NULL | |
| SemNo | char(1) | YES | | NULL | |
| HallNo | varchar(4) | YES | | NULL | |
| FacNo | varchar(4) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)

mysql> ALTER TABLE FACULTY
-> ADD Dept VARCHAR(10);
Query OK, 0 rows affected (0.08 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> desc faculty;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| FacNo | varchar(4) | NO | PRI | NULL | |
| FacName | varchar(15) | YES | | NULL | |
| Gender | char(1) | YES | | NULL | |
| DOB | date | YES | | NULL | |
| DOJ | date | YES | | NULL | |
| MobileNo | bigint | YES | | NULL | |
| Dept | varchar(10) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
7 rows in set (0.00 sec)

mysql>
```

DML

1)

```
MySQL 8.0 Command Line Client
-> DOB DATE,
-> MobileNo BIGINT,
-> City VARCHAR(15)
-> );
ERROR 1046 (3D000): No database selected
mysql> use dbms;
Database changed
mysql> CREATE TABLE STUDENT2
-> (
-> RegNo INT,
-> SName VARCHAR(20),
-> Gender CHAR(1),
-> DOB DATE,
-> MobileNo BIGINT,
-> City VARCHAR(15)
-> );
Query OK, 0 rows affected (0.01 sec)

mysql> ALTER TABLE STUDENT2
-> ADD PRIMARY KEY (RegNo);
Query OK, 0 rows affected (0.05 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> desc student2;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| RegNo | int | NO | PRI | NULL | |
| SName | varchar(20) | YES | | NULL | |
| Gender | char(1) | YES | | NULL | |
| DOB | date | YES | | NULL | |
| MobileNo | bigint | YES | | NULL | |
| City | varchar(15) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)

mysql>
```

2)

```
MySQL 8.0 Command Line Client

mysql> ALTER TABLE FACULTY
-> ADD PRIMARY KEY (FacNo);
Query OK, 0 rows affected (0.04 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> desc faculty;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| FacNo  | varchar(4)    | NO   | PRI | NULL    |       |
| FacName | varchar(15)   | YES  |     | NULL    |       |
| Gender | char(1)       | YES  |     | NULL    |       |
| DOB    | date         | YES  |     | NULL    |       |
| DOJ    | date         | YES  |     | NULL    |       |
| MobileNo | bigint       | YES  |     | NULL    |       |
| Dept   | varchar(10)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
7 rows in set (0.00 sec)

mysql>
```

```
MySQL 8.0 Command Line Client

mysql> ALTER TABLE FACULTY
-> ADD PRIMARY KEY (FacNo);
Query OK, 0 rows affected (0.04 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> desc faculty;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| FacNo | varchar(4) | NO | PRI | NULL | |
| FacName | varchar(15) | YES | | NULL | |
| Gender | char(1) | YES | | NULL | |
| DOB | date | YES | | NULL | |
| DOJ | date | YES | | NULL | |
| MobileNo | bigint | YES | | NULL | |
| Dept | varchar(10) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
7 rows in set (0.00 sec)

mysql> ALTER TABLE FACULTY
-> ADD CONSTRAINT chk_gender
-> CHECK (Gender IN ('M','F'));
Query OK, 0 rows affected (0.06 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> desc faculty;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| FacNo | varchar(4) | NO | PRI | NULL | |
| FacName | varchar(15) | YES | | NULL | |
| Gender | char(1) | YES | | NULL | |
| DOB | date | YES | | NULL | |
| DOJ | date | YES | | NULL | |
| MobileNo | bigint | YES | | NULL | |
| Dept | varchar(10) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
7 rows in set (0.00 sec)

mysql>
```

3)

```
mysql> ALTER TABLE DEPARTMENT
-> DROP PRIMARY KEY;
Query OK, 0 rows affected (0.06 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> ALTER TABLE DEPARTMENT
-> ADD PRIMARY KEY (DeptNo);
Query OK, 0 rows affected (0.04 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> desc department;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| DeptNo | varchar(4) | NO | PRI | NULL | |
| DeptName | varchar(15) | YES | | NULL | |
| DeptHead | varchar(4) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql>
```

4)

```
mysql> alter table course
    -> drop primary key;
Query OK, 0 rows affected (0.09 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> desc course;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| CourseNo | varchar(3) | NO | | NULL | |
| CourseDesc | varchar(14) | YES | | NULL | |
| CourseType | char(1) | YES | | NULL | |
| SemNo | char(1) | YES | | NULL | |
| HallNo | varchar(4) | YES | | NULL | |
| FacNo | varchar(4) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)

mysql> ALTER TABLE COURSE
    -> ADD PRIMARY KEY (CourseNo);
Query OK, 0 rows affected (0.04 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> desc course;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| CourseNo | varchar(3) | NO | PRI | NULL | |
| CourseDesc | varchar(14) | YES | | NULL | |
| CourseType | char(1) | YES | | NULL | |
| SemNo | char(1) | YES | | NULL | |
| HallNo | varchar(4) | YES | | NULL | |
| FacNo | varchar(4) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)

mysql>
```

```
MySQL 8.0 Command Line Client
+-----+-----+-----+-----+-----+-----+
| FacNo | varchar(4) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)

mysql> ALTER TABLE COURSE
    -> ADD CONSTRAINT chk_sem
    -> CHECK (SemNo IN ('1','2','3','4','5','6'));
Query OK, 0 rows affected (0.06 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> desc course;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| CourseNo | varchar(3) | NO | PRI | NULL | |
| CourseDesc | varchar(14) | YES | | NULL | |
| CourseType | char(1) | YES | | NULL | |
| SemNo | char(1) | YES | | NULL | |
| HallNo | varchar(4) | YES | | NULL | |
| FacNo | varchar(4) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)

mysql>
```

5)

```
6 rows in set (0.00 sec)

mysql> ALTER TABLE STUDENT2
    -> ADD CONSTRAINT uq_sname
    -> UNIQUE (SName);
Query OK, 0 rows affected (0.02 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> desc student2
    -> ;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| RegNo | int  | NO   | PRI | NULL    |       |
| SName | varchar(20) | YES | UNI | NULL    |       |
| Gender | char(1) | YES |     | NULL    |       |
| DOB   | date  | YES |     | NULL    |       |
| MobileNo | bigint | YES |     | NULL    |       |
| City  | varchar(15) | YES |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)

mysql> 
```

6)

```
MySQL 8.0 Command Line Client

mysql> CREATE TABLE STUDENT3
    -> (
    ->   RegNo INT,
    ->   Dept VARCHAR(15)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> ALTER TABLE STUDENT2
    -> ADD PRIMARY KEY (RegNo);
ERROR 1068 (42000): Multiple primary key defined
mysql> ALTER TABLE STUDENT3
    -> ADD CONSTRAINT FK_REGNO
    -> FOREIGN KEY (RegNo)
    -> REFERENCES STUDENT2(RegNo);
Query OK, 0 rows affected (0.08 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> DESC STUDENT2;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| RegNo | int  | NO   | PRI | NULL    |       |
| SName | varchar(20) | YES | UNI | NULL    |       |
| Gender | char(1) | YES |     | NULL    |       |
| DOB   | date  | YES |     | NULL    |       |
| MobileNo | bigint | YES |     | NULL    |       |
| City  | varchar(15) | YES |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)

mysql> DESC STUDENT3;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| RegNo | int  | YES  | MUL | NULL    |       |
| Dept  | varchar(15) | YES |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> 
```

MySQL 8.0 Command Line Client

```
-> FOREIGN KEY (RegNo)
-> REFERENCES STUDENT2(RegNo);
Query OK, 0 rows affected (0.08 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

mysql> DESC STUDENT2;

Field	Type	Null	Key	Default	Extra
RegNo	int	NO	PRY	NULL	
SName	varchar(20)	YES	UNI	NULL	
Gender	char(1)	YES		NULL	
DOB	date	YES		NULL	
MobileNo	bigint	YES		NULL	
City	varchar(15)	YES		NULL	

6 rows in set (0.00 sec)

mysql> DESC STUDENT3;

Field	Type	Null	Key	Default	Extra
RegNo	int	YES	MUL	NULL	
Dept	varchar(15)	YES		NULL	

2 rows in set (0.00 sec)

mysql> SHOW CREATE TABLE STUDENT3;

Table	Create Table
STUDENT3	CREATE TABLE `student3` (`RegNo` int DEFAULT NULL, `Dept` varchar(15) DEFAULT NULL, KEY `FK_REGNO` (`RegNo`), CONSTRAINT `FK_REGNO` FOREIGN KEY (`RegNo`) REFERENCES `student2` (`RegNo`)) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci

1 row in set (0.00 sec)